Inverse z-Transforms

Rational z-Transform

$$X(z) = \frac{P(z)}{Q(z)}$$

$$= \frac{a_0 z^N + a_1 z^{N-1} + \cdots + a_N}{b_0 z^M + b_1 z^{M-1} + \cdots + b_M}$$

ROC R

degree Numerator = N $\quad a_0 \neq 0$

degree of denom. = M $\quad b_0 \neq 0$

Assume $b_0 = 1$

$$X(z) = \frac{z^N (a_0 + a_1 z^{-1} + \cdots + a_N z^{-N})}{z^M (1 + b_1 z^{-1} + \cdots + b_M z^{-M})}$$

$z^{-(M-N)}$

resulting time shift of $(M-N)$

Let us invert

$$X_1(z) = \frac{a_0 + a_1 z^{-1} + \cdots + a_N z^{-N}}{1 + b_1 z^{-1} + \cdots + b_M z^{-M}}$$

ROC R

deg Num < deg denm

Case A $N < M$

Case A1 Subcase: All M roots of denominator are all distinct

$$X_1(z) = \frac{P_1(z)}{Q_1(z)}$$

$$Q_1(z) = 1 + b_1 z^{-1} + \cdots + b_M z^{-M}$$

Roots of $Q(z)$ are distinct.

$p_1, p_2, \ldots, p_M$ be roots of $Q_1(z)$

$$Q_1(z) = (1 - p_1 z^{-1})(1 - p_2 z^{-1}) \cdots (1 - p_M z^{-1})$$

Partial fraction

$$\frac{a_0 + a_1 z^{-1} + \cdots + a_M z^{-M}}{(1 - p_1 z^{-1})(\cdots)(1 - p_M z^{-1})} = X_1(z) = \frac{C_1}{1 - p_1 z^{-1}} + \frac{C_2}{1 - p_2 z^{-1}} + \cdots + \frac{C_M}{1 - p_M z^{-1}}$$

$$X_1(z)(1 - p_1 z^{-1}) = C_1 + C_2 \frac{(1 - p_1 z^{-1})}{(1 - p_2 z^{-1})} + \cdots + C_M \frac{(1 - p_1 z^{-1})}{(1 - p_M z^{-1})}$$

(the C_2 term is 0 @ $z = p_1$; the C_M term is 0)

@ $z = p_1$

$$= C_1 = X_1(z)(1 - p_1 z^{-1}) \Big|_{@ z = p_1}$$

$$C_k = X_1(z)(1 - p_k z^{-1}) \Big|_{@ z = p_k}$$

(causal) $c_1 p_1^n u(n) \Rightarrow$ $\quad \Rightarrow -c_M p_M^n u(-n-1)$ (anti-causal)

$$X_1(z) = \frac{c_1}{1-p_1 z^{-1}} + \cdots + \frac{c_M}{1-p_M z^{-1}} \qquad \text{ROC } R$$

There are $M+1$ Possible Rocs

Suppose $|p_1| < |p_2| < |p_3| < \cdots < |p_M|$

Possible Roc's

ROC R_1 : $|z| < |p_1|$

R_2 : $|p_1| < |z| < |p_2|$ $\Rightarrow$ Roc is outside p_1 (p_1 causal)
Roc inside $p_2, p_3, \cdots, p_M$ anticausal.

R_M : $|p_{M-1}| < |z| < |p_M|$

R_{M+1} : $|z| > |p_M|$

Case A2

$$X_1(z) = \frac{a_0 + a_1 z^{-1} + \cdots + a_N z^{-N}}{1 + b_1 z^{-1} + \cdots + b_M z^{-M}}$$

de $N < M$

$Q_1(z) = 1 + b_1 z^{-1} + \cdots + b_M z^{-M}$ $\Rightarrow$ Has multiplicity of roots

Example $\quad Q_1(z) = (1-p_1 z^{-1})^2 (1-p_2 z^{-1})$

Pole at p_1 has multiplicity 2

In this case

$$X_1(z) = \frac{C_{11}}{1-p_1 z^{-1}} + \frac{C_{12}}{(1-p_1 z^{-1})^2} + \frac{C_2}{(1-p_2 z^{-1})}$$

(differentiation property)

Causal inverse $\Rightarrow C_{12}(n+1)\, p_1^n\, u(n)$

anticausal inverse $\Rightarrow$ find yourself

$$C_2 = X_1(z)\,(1-p_2 z^{-1})\,\Big|_{@\, z=p_2}$$

$$C_{12} = X_1(z)\,(1-p_1 z^{-1})^2\,\Big|_{@\, z=p_1}$$

$$\text{LHS} = X_1(z)\Big|_{z=\alpha} = \frac{C_{11}}{1-p_1/\alpha} + \frac{C_{12}}{(1-(p_1/\alpha))^2} + \frac{C_2}{1-p_2/\alpha}$$

C_{11}, C_{12}, C_2 known

$\alpha \neq p_1, p_2$

Solve for C_{11}

Case B

$$X(z) = \frac{P(z)}{Q(z)}$$

degree N / degree M

degree $(P(z))$ $\geq$ degree of $Q(z)$

$$X(z) = \underbrace{\alpha_0 z^0 + \alpha_1 z^{-1} \cdots \alpha_{N-M} z^{-(N-M)}}_{A(z),\ \text{degree } (N-M)} + \underbrace{\frac{B(z)}{Q(z)}}_{\text{degree} < M \text{ / degree } M}$$

Previous case

$A(z) \Rightarrow$ quotient when $P(z)$ is divided by $Q(z)$

$B(z) \Rightarrow$ Remainder " $P(z)$ " $Q(z)$

Long Division of Polynomials.

$$X(z) = \frac{1}{1 - az^{-1}}$$

$$= \frac{z}{z - a}$$

Two Possible ROC

Case 1: $|z| > |a|$ causal

Case 2: $|z| < |a|$ anti-causal

$|z| > |a|$ causal

Case 1

$\Rightarrow a^n u(n)$

$$z - a \;\Big|\; z \qquad 1 + az^{-1} + a^2z^{-2} + \dots$$

$z - a$

a

$a - a^2z^{-1}$

a^2z^{-1}

$a^2z^{-1} - a^3z^{-2}$

$\vdots$

Case 2 $|z| < |a|$ anticausal

$$-a + z \;\Big|\; z \qquad -\tfrac{1}{a}z - \tfrac{1}{a^2}z^2 - \dots$$

$z - \frac{1}{a}z^2$

$\frac{1}{a}z^2$

$\frac{1}{a}z^2 - \frac{1}{a^2}z^3$

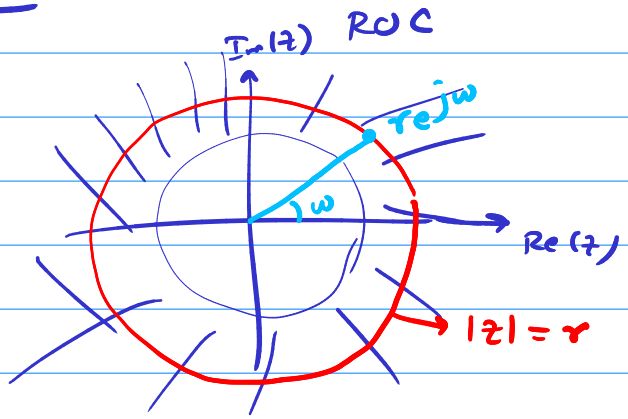
$\dots$

Integration in Complex plane

$X(z)$ has $|z| = r$ is in ROC

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

Consider $z = re^{j\omega}$



$$X(z)\Big|_{@\, z = re^{j\omega}} = X(re^{j\omega})$$

$$= \sum_n x(n) \left(re^{j\omega}\right)^{-n}$$

$$X(re^{j\omega}) = \sum_n x(n) r^{-n} e^{-j\omega n}$$

$\omega \Rightarrow 0$ to 2π

$r \Rightarrow$ fixed

$\omega \Rightarrow$ Variable

$$X(re^{j\omega}) = DTFT\left(x(n) r^{-n}\right)$$

Inverse DTFT

$$x(n) r^{-n} = \frac{1}{2\pi} \int_{\omega=0}^{2\pi} X(re^{j\omega}) e^{j\omega n} \, d\omega$$

$$\boxed{x(n) = \frac{1}{2\pi} \int_{\omega=0}^{2\pi} X(re^{j\omega}) r^n e^{j\omega n} \, d\omega}$$

r is fixed

Power series method (irrational)

$$X(z) = \log\left(1 + az^{-1}\right) \; ; \quad |z| > |a|$$

$$\log(1+t) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{t^n}{n} \; ; \quad |t| < 1$$

take $t = az^{-1}$

$|t| < 1$ if $|z| > |a|$

$$X(z) = \log(1 + az^{-1}) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{(az^{-1})^n}{n}$$

$$x(n) = \begin{cases} (-1)^{n+1} \frac{a^n}{n} & ; \ n \geq 1 \\ 0 & ; \ \text{else} \end{cases}$$

x ——————— x

Initial Value Theorem

- $x(n)$ is a causal signal

$$X(z) = x(0) + x(1) z^{-1} + x(2) z^{-2} + \cdots$$

$$\lim_{z \to \infty} x(z) = x(0)$$

Final Value Theorem

$x(n)$ is a causal signal

$$\lim_{z \to 1} X(z)(1 - z^{-1}) = x(\infty) = \lim_{N \to \infty} x(N)$$

Provided the limits exist.

Proof.

$$v(n) = x(n) - x(n-1)$$

$$V(z) = X(z) - z^{-1} x(z) = (1 - z^{-1}) X(z)$$

$$\lim_{z \to 1} V(z) = \lim_{z \to 1} \sum_{n=0}^{\infty} v(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} v(n) 1^{-n}$$

$$= \lim_{N\to\infty} \sum_{n=0}^{N} v(0) + v(1) + v(2) + \cdots$$

telescoping sum

$$= x(0) - x(-1) + x(1) - x(0) + x(2) - x(1) + \cdots$$

$x(-1) = 0$

$$= \lim_{N\to\infty} x(N) = x(\infty)$$



LCCDE $\quad y \to$ output $\quad x \to$ input

$$b_0 y(n) + b_1 y(n-1) + \cdots + b_M y(n-M)$$

$$= a_0 x(n) + \cdots + a_N x(n-N)$$

Assume zero initial condition $\Rightarrow$ LTI system

delay property

$$b_0 Y(z) + b_1 z^{-1} Y(z) + \cdots + b_M z^{-M} Y(z)$$

$$= a_0 X(z) + \cdots + a_N z^{-N} X(z)$$

Transfer function

$$\frac{Y(z)}{X(z)} = H(z) = \frac{a_0 + a_1 z^{-1} + \cdots + a_N z^{-N}}{b_0 + b_1 z^{-1} + \cdots + b_M z^{-M}}$$

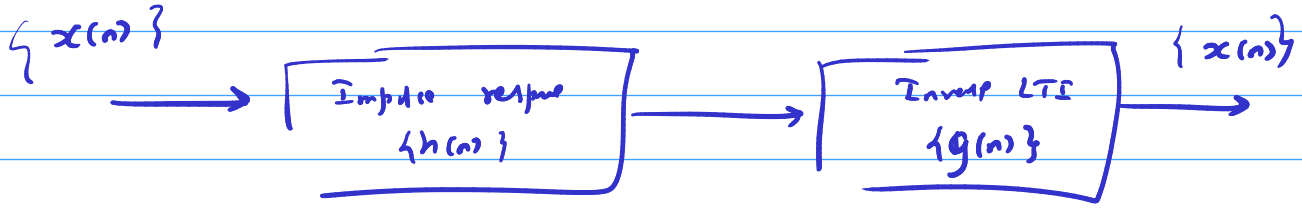
$=$ rational transfer function

To find impulse response we need ROC information.

For instance if system causal, ROC lies outside largest pole



Inverse of LTI System



$$\{h(n)\} * \{g(n)\} = \{\delta(n)\}$$

In z domain

$$H(z) \cdot G(z) = 1$$

$$G(z) = \frac{1}{H(z)}$$

Poles of H become zeros of G

Zeros of H become poles of G.

Suppose H is causal & stable. All poles should be inside unit circle. & ROC lies outside largest pole.

For inverse to be causal & stable

Poles of G (zeros of H) should lie unit circle.